

## 7. Capacitors

Storing energy in an electric field

**Lesson 7** · about 90 minutes · everyone builds; battery power only

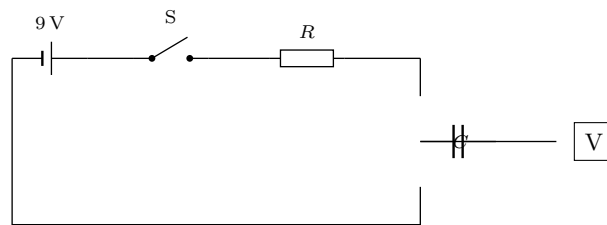
### What you need

- Capacitors: 100  $\mu\text{F}$ , 470  $\mu\text{F}$ , 1000  $\mu\text{F}$  electrolytic
- Resistors: 1 k $\Omega$ , 10 k $\Omega$ , 100 k $\Omega$
- LED, breadboard, 9 V battery, switch
- Multimeter, stopwatch
- A plastic film canister or small jar, aluminium foil, tape
- The electroscope from Lesson 1

### Predict first

A 1000  $\mu\text{F}$  capacitor is charged from a 9 V battery through a 10 k $\Omega$  resistor. How long until it reaches about two-thirds of 9 V? Then: you disconnect the battery entirely and short the capacitor's legs with a screwdriver a minute later. Does anything happen?

### Do it



- Charge and watch.** Build the circuit with 1000  $\mu\text{F}$  and 10 k $\Omega$ . Close the switch and read the voltage every two seconds. It rises fast, then slower, and creeps toward 9 V without ever quite arriving.
- Discharge.** Open the switch and connect a resistor across the capacitor. Watch it fall the same way, mirrored.
- Change one thing.** Repeat with 100 k $\Omega$ , then back to 10 k $\Omega$  but with 100  $\mu\text{F}$ . Time how long each takes to reach about two-thirds. Multiply  $R$  by  $C$  each time and compare.
- Store enough to see.** Charge 1000  $\mu\text{F}$  to 9 V, disconnect the battery, then connect an LED across it. It glows and fades. The battery is gone — that light is coming out of the capacitor.
- Build a Leyden jar.** Wrap the outside of a film canister in foil and line the inside with foil, keeping the two layers from touching. Push a wire through the lid to the inner foil. Charge it from a comb rubbed on wool, many strokes, holding the outer foil. Then bring the wire near the outer foil — with a *dry stick*, not your fingers — and watch the spark.
- Feel the difference.** A 1000  $\mu\text{F}$  capacitor at 9 V is harmless. The Leyden jar holds far less charge at a far higher voltage and it will make a spark and sting. Same energy scale, entirely different behaviour.

### What it means

|                                     |                                                                                                                                                                                                                     |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>What a capacitor is</b>          | Two conductors separated by an insulator. Charge piles up on one plate and is repelled from the other, and the energy is stored in the electric field <i>between</i> them — the field you made visible in Lesson 2. |
| <b>Capacitance</b>                  | $Q = CV$ . Bigger plates and a thinner gap store more charge for the same voltage.                                                                                                                                  |
| <b>The time constant</b>            | $\tau = RC$ . In one time constant the capacitor charges to about 63% of the way, and after five it is essentially there. Your three timings should have scaled exactly with the product $R \times C$ .             |
| <b>Stored energy</b>                | $E = \frac{1}{2}CV^2$ . Note the <i>square</i> : doubling the voltage quadruples the stored energy. This is the single most important equation in this course, and Lesson 11 is built on it.                        |
| <b>A capacitor is not a battery</b> | It stores far less energy, but it can release all of it in microseconds. A battery is a reservoir; a capacitor is a bucket you can tip over instantly. A Tesla coil needs the tipping.                              |

### THE HABIT THAT MATTERS LATER

The Leyden jar holds its charge after you stop rubbing, and a capacitor bank in Lesson 11 holds a lethal charge after the power is off. Practise the habit here, where the worst case is a sting: **before you touch a capacitor, short it out with something you are holding by an insulated handle.** Then short it again. Large capacitors can partially recharge themselves through dielectric absorption after being discharged once. This is exactly why the grounding stick exists.

### The link to the coil

The tank capacitor is the shot the coil fires. The transformer charges it over several thousandths of a second; the spark gap dumps it in a few millionths.  $E = \frac{1}{2}CV^2$  says how much energy is in that shot, and because power is energy divided by time, dumping a modest amount of energy in a very short time produces an enormous power — which is the whole trick.

**This lesson is low voltage.** Everything here runs from batteries or a small plug-in supply and cannot hurt you. That changes at Lesson 10. Build the habits now — one hand, discharge before touching, check before you power up — while the stakes are nothing, so that they are automatic when the stakes are real.

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